

ROOTMESH · DESIGN DECISION RECORD

RootMesh recommendations accepted

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DOCUMENT CONTROL

Document ID	RMS-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/rootmesh		
Source	docs/decisions/0002-recommendations-accepted.md @ 078467c		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of the remaining review recommendations and what changed in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted for items A1 to A4; items O1 and P1 remain proposed

■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." The TRL 2 review items that carried a recommendation (D1 to D9) had already been decided in RMS-DDR-001 and are unchanged. This record covers the recommendations that the review note (docs/REVIEW.md) and the TRL 3 documents made after that: the two suggestions in the TRL 2 review note and the recommended next step in the TRL 3 review note, which proposed design responses to the at-risk requirements R9 (cell temperature) and R10 (installation force). Items without a recommendation stay open. TRL 4 remains on hold by Amish's instruction, so any recommendation that needs building, testing or measuring is recorded as decided but on hold.

■ Decision

Table 1. Newly decided items.

#	Item	Decision	What changed in the repo
A1	Slot tool for firm soil (TRL 2 review, suggestion; RMS-CAL-001 v0.1, section G)	Decided by Amish, 2026-09-25: go with recommendation. A tool pre-cuts the fin path. At TRL 3 it is a 32 x 8 mm mild steel flat bar, 650 mm, pointed at the EC tip depth, with a bolted T-handle and a printed depth-stop collar; steel rather than printed, because a printed blade would not survive firm soil.	cad/src/model.py adds build_slot_tool() and exports slot-tool.step and .stl ; BOM line 15 (\$10.00, one per set); RMS-CAL-001 v0.2 section G: push force 517 N to 376 N in moist loam and 1,778 N to 1,216 N in firm dry loam; R10 stays at risk; RMS-DWG-001 Rev P1 to P2 (slot tool note); RMS-PRC-001 v0.4 (component 15, design choice, safety).
A2	Deeper cell position for R9 (TRL 3 review, recommended next step)	Decided by Amish, 2026-09-25: go with recommendation. The deeper cell was checked by calculation.	RMS-CAL-001 v0.2 [F4]: 40 °C needs the cell center about 163 mm deep, where the upper probe sits; the deepest center that fits is about 65 mm (46.5 °C against 46.8 °C). The cell stays at 62 mm; no geometry change. R9 stays at risk and the remaining choice becomes P1 below.
A3	Pilot set now includes the slot tool (follows from A1)	Decided by Amish, 2026-09-25: go with recommendation. budget_usd stays \$300.	R12 restated so that the \$300 pilot set covers three stakes, one gateway and the slot tool; pilot set \$256.50 to \$266.50, headroom \$43.50 to \$33.50 (RMS-REQ-001 v0.4; bom/bom-notes.md ; concept blueprint key figures).
A4	Calibration note for the two-point field calibration (TRL 2 review, suggestion)	Decided by Amish, 2026-09-25: go with recommendation, but on hold. The EC cell constant is already calculated in RMS-CAL-001 section D; the field moisture calibration needs gravimetric samples and KCl standards, which is TRL 4 work.	None. On hold because TRL 4 is on hold by Amish's instruction.

Table 2. Items still open.

#	Item	Status
O1	First users and region for co-design (market garden, orchard, research farm or extension program), which also sets EU868 or US915. No recommendation was made.	Proposed, awaiting Amish
P1	Response to the cell temperature in bare, hot soil (R9), raised by the check in A2. Option A: a printed shade skirt around the head to shade the soil over the cell (adds a part near machinery; its effect needs measuring). Option B: restate the R9 cell limit to the chosen cell's rated discharge and storage range, taken from its datasheet, while charging stays blocked above 45 °C. Recommendation: B, because the charger already blocks charging in the hot hours and B adds no part; confirm the cell's rated range before adopting it. This is a new proposal made after Amish's acceptance, so it is not decided.	Proposed, awaiting Amish

■ Consequences

- `project.yaml`: `budget_usd` stays \$300; pitch and problem unchanged; `trl: 3` and `trl_target: 3`.
- RMS-PRC-001 and RMS-REQ-001 move to v0.4, RMS-CAL-001 to v0.2, and drawing RMS-DWG-001 to Rev P2. RMS-PRB-001 is unchanged by these decisions.
- Requirement status is unchanged in count: 8 met, 5 at risk (R1, R5, R9, R10, R14), 1 not verifiable at TRL 3 (R8), none not met. R10 improves in moist soil but stays at risk in firm dry soil.
- No cross-repo action follows from these items.
- Nothing here moves the design toward TRL 4. A timed installation trial, the field calibration and any build of the slot tool are TRL 4 work and on hold.